

# Holden Swindell

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|-------------|-------------------------------------------------------------------------------------------------------|
| Email       | swind027@umn.edu                                                                                      |
| ORCID       | 0009-0002-9619-9386  |
| Website     | <a href="https://hswindell.github.io">https://hswindell.github.io</a>                                 |
| Institution | University of Minnesota Twin Cities                                                                   |

## Education

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| University of Minnesota Twin Cities | August 2025 – Present |
| <i>Mathematics, PhD</i>             |                       |

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| University of Wisconsin-Madison                            | September 2021 – May 2025 |
| <i>Mathematics, BSc and Computer Sciences, Certificate</i> |                           |

## Research Experience

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|-----------------------------------------------------------------------------------------|-------------|
| UW-Madison REU in Analysis                                                              | Summer 2023 |
| <i>Analysis and Partial Differential Equations RTG, University of Wisconsin-Madison</i> |             |

Website: <https://web.math.wisc.edu/analysisrtg/node/7>

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| MXM Project “Random Symmetries of Hyperbolic Spaces”                               | Fall 2022 |
| <i>Madison Experimental Mathematics (MXM) Lab, University of Wisconsin-Madison</i> |           |

Website: <https://mxm.math.wisc.edu/past-semesters/fall-2022>

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|------------------------------------------------------------------------------|-------------|
| MXM Project “Geometry of Random Integer Matrices”                            | Spring 2022 |
| <i>Madison Experimental Mathematics Lab, University of Wisconsin-Madison</i> |             |

Website: <https://mxm.math.wisc.edu/past-semesters/sp22>

## Teaching Experience

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|            |                         |
|------------|-------------------------|
| Lead Tutor | Fall 2024 – Spring 2025 |
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| Program Assistant | Summer 2024 |
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| Lead Tutor | Spring 2024 |
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| Tutor | Spring 2023 – Fall 2023 |
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| Tutor                                                        | Spring 2022 – Fall 2022 |
| <i>Math Learning Center, University of Wisconsin-Madison</i> |                         |

Website: <https://math.wisc.edu/undergraduate/mlc>

## Service & Community Outreach

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| President | Fall 2024 – Spring 2025 |
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| Vice President | Fall 2023 – Spring 2024 |
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| Board Advisor                                     | Fall 2022 – Spring 2023 |
| <i>Math Club, University of Wisconsin-Madison</i> |                         |

Website: <https://win.wisc.edu/organization/mathclub>

## Directed Reading Mentor

Spring 2024

*Category Theory Reading Group, University of Wisconsin-Madison*

Website: <https://hswindell.github.io/category-theory.html>

## Honors & Awards

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UW-MADISON

Honors in the Liberal Arts  
Honors in the Major (Mathematics)  
Phi Beta Kappa Honor Society Member  
Dean's List (all Semesters)

## Technical Skills

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PROGRAMMING

C, Python, Rust

WEB

HTML, CSS

DOCUMENT PREPARATION

L<sup>A</sup>T<sub>E</sub>X, Microsoft Office Suite, Google Docs Editors Suite

## Talks

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**“Point-Free Topology”**

April 2024

*Category Theory Reading Group, University of Wisconsin-Madison*

**“Smooth Infinitesimal Analysis”**

February 2023

*Math Club, University of Wisconsin-Madison*